

# Which applications should a review team see first?

*A cost-aware review policy for synthetic financing applications, under a fixed 12% review capacity*

**7.89%**

of 10,000 training applications default  
within 90 days

**10 : 1**

cost of a missed default vs. an  
unnecessary review

**12%**

maximum share of applications  
reviewed in each period

**Decision supported:** flag an application for human review (1) or not (0). A flag is not a refusal, and 0 is not an approval.

**Target:** synthetic default within 90 days after the application date.

**Success means:** lowest educational loss ( $10 \cdot \text{FN} + \text{FP}$ ) among thresholds that stay within 12% in every period; ranking judged by Average Precision.

# A method that keeps the future out of training

1

## Data

Leakage audit removed 2 post-outcome columns; IDs and dates excluded

2

## Forward OOF

Train before validate, customers kept apart, 90-day labels mature first

3

## Selection

Worth-It Gate: ensemble must beat best single by > 1 fold SD

4

## Calibration

Sigmoid fitted on a reserved later period, never on selection rows

5

## Policy

Threshold chosen on OOF; 12% cap applied once to the whole batch

### Why it matters: the same model, two validations

0.999

AP with leaked columns, random split

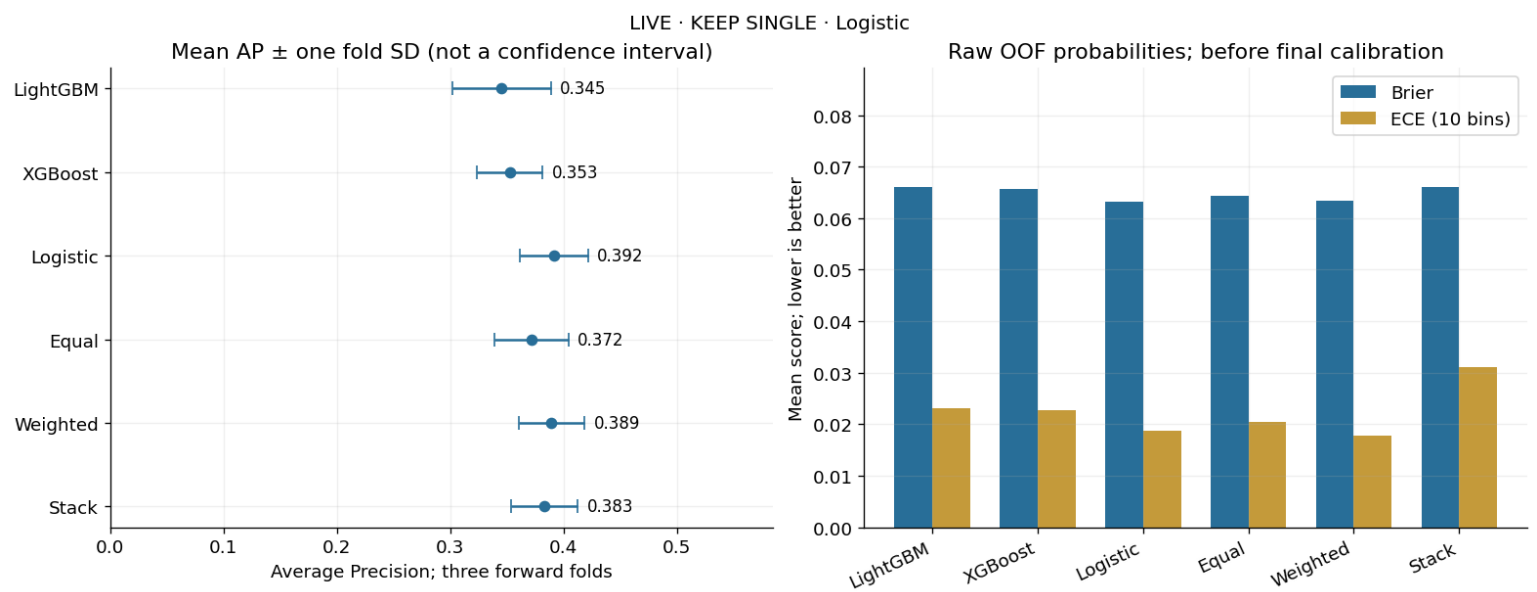
0.315

AP with honest time + customer split

### Where each piece was learned

- Model and imputers: training folds only
- Ensemble weights and stacker: inner OOF only
- Calibrator: reserved calibration period
- Threshold: OOF predictions, never the challenge data

# Ensembles did not earn their cost: keep Logistic Regression



| Candidate         | Mean AP | vs. Logistic |
|-------------------|---------|--------------|
| Logistic (chosen) | 0.392   | —            |
| Weighted ensemble | 0.389   | −0.002       |
| Stacking          | 0.383   | −0.009       |
| Equal average     | 0.372   | −0.020       |

Gate: lift must exceed one fold SD (0.030). Fold SD is descriptive, from three overlapping folds.

**Decision: KEEP SINGLE**

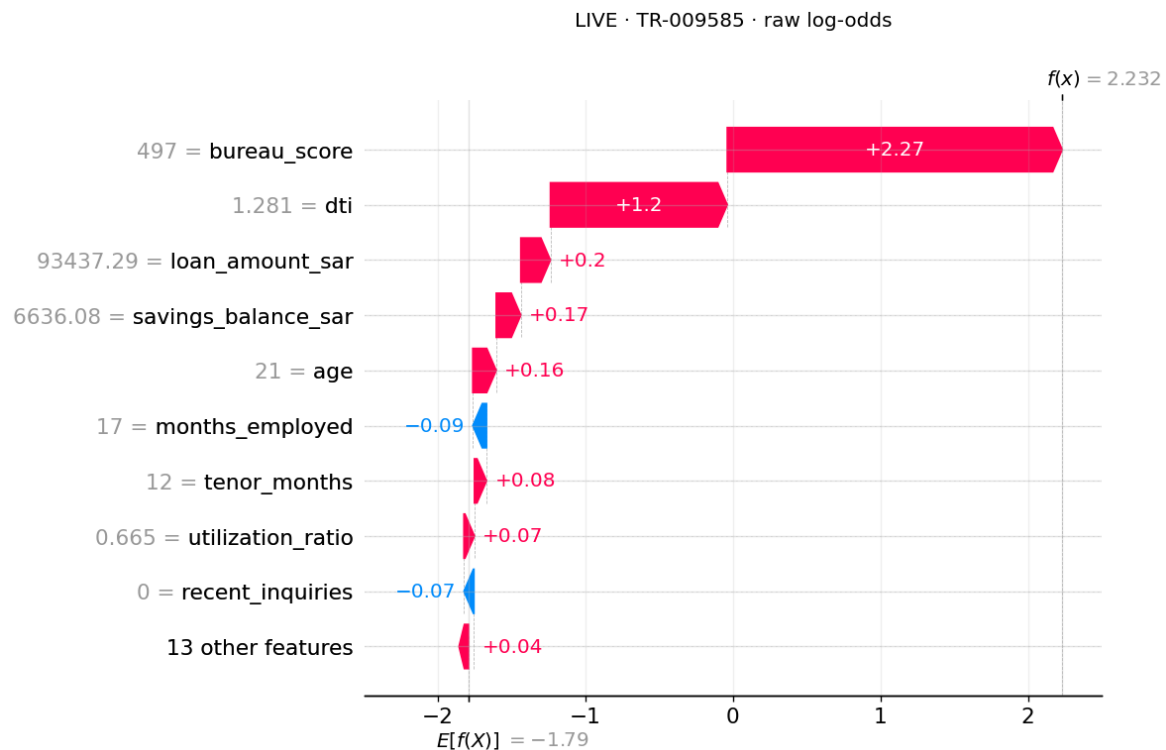
Policy evidence (OOF, 2,155 rows)

46.9%

of defaults caught (84 of 179); precision 34.3%

- Busiest period: 11.7% flagged ( $\leq 12\%$ )
- Loss 1,111 units — optimistic, chosen on the same OOF
- Challenge: 330 passed the threshold, cap kept the top 300
- No challenge performance claimed: no labels

# Explanations and probabilities, with clear limits



Day 4 weighted LightGBM, raw log-odds. Not the final Logistic model.

## One high-score case (TR-009585)

bureau\_score 497 (+2.27) and dti 1.28 (+1.20) drove the score: associations in the model, not causes of default

| Calibration (ECE) | Raw   | Sigmoid | Evidence        |
|-------------------|-------|---------|-----------------|
| Day 4 LightGBM    | 0.147 | 0.023   | separate period |
| Day 5 Logistic    | 0.021 | 0.035   | fit rows only   |

## What this means

- Class weighting inflated LightGBM scores, so the sigmoid helped a lot
- Logistic was already calibrated; the sigmoid slightly hurt but keeps the order, so the same 300 are flagged
- Still to check: contributions on the final model, and calibration once 90-day labels mature

# Ship the simple model, and watch capacity

*Logistic Regression ranks applications for review, catching about half of defaults inside the 12% cap; ensembles added cost without evidence of gain.*

## Top two limitations

- Development evidence only: OOF on reused course data; loss is optimistic; no challenge labels
- Capacity pressure: challenge batch was 13.2% before the cap; Day 4 Q4 also exceeded capacity

## Follow-up plan

- Track each batch's pre-cap flag rate against 12%
- Recheck Brier, ECE and AP when 90-day outcomes mature
- Watch score drift and regional false-alarm rates (western 10.3% vs eastern 6.3%)

## What would change it

- An ensemble beating Logistic by more than one fold SD on new folds
- Pre-cap flags staying above 12%: redevelop the threshold on new data
- Sigmoid hurting again on matured labels: remove it via a governed update

Repository: [FelwaAl/FulwahAlarifi\\_Tamweel-Lite\\_AdvancedMachineLearning\\_211](#): Final project of SADAIA's program (Advanced Machine Learning) class 211.

• Final tag: v1.0-final